

PROJECT REPORT: ONLINE SHOPPING DATABASE SYSTEM

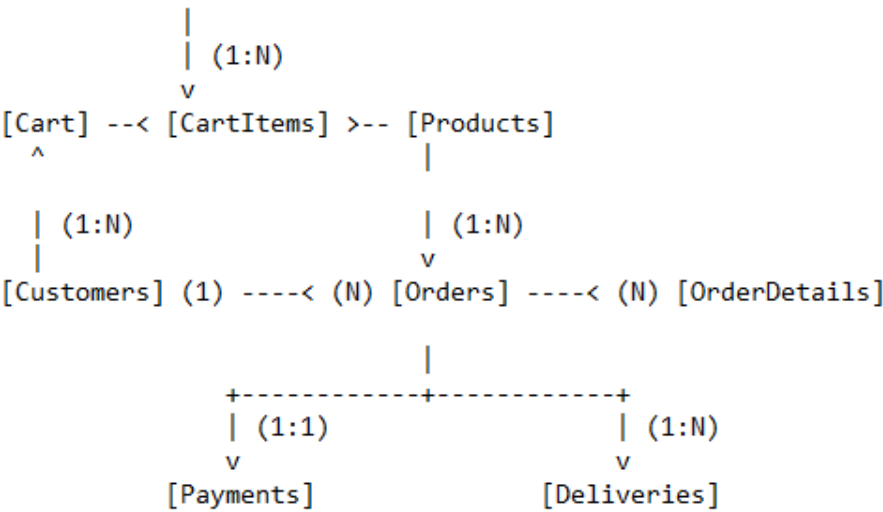
Technical Architecture, ERD, and Security Implementation

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REG NO: 2025/BCS/021/PS
PROJECT NAME: ONLINE SHOPPING
BUSINESS CASE: E-COMMERCE
INDIVIDUAL PROJECT: SECURITY AND ACCESS CONTROL

1. Executive Summary

This project involved the design and development of an Online Shopping Database. The system is designed to manage a full e-commerce lifecycle including inventory management, customer authentication, and order fulfillment. My specialized contribution focused on **Security and Access Control**, ensuring data privacy via cryptographic hashing and protecting system integrity through Role-Based Access Control (RBAC).

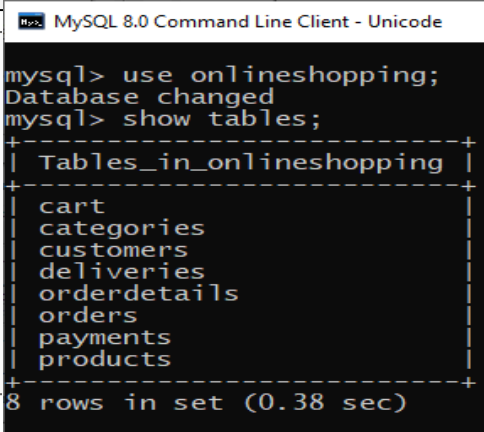
[Categories]



3. Database Structure (DDL)

The database structure was implemented using SQL Data Definition Language (DDL) to create normalized tables with appropriate indexing and relational integrity.

```
mysql> CREATE DATABASE OnlineShopping;
mysql> USE OnlineShopping;
mysql> SHOW TABLES;
```

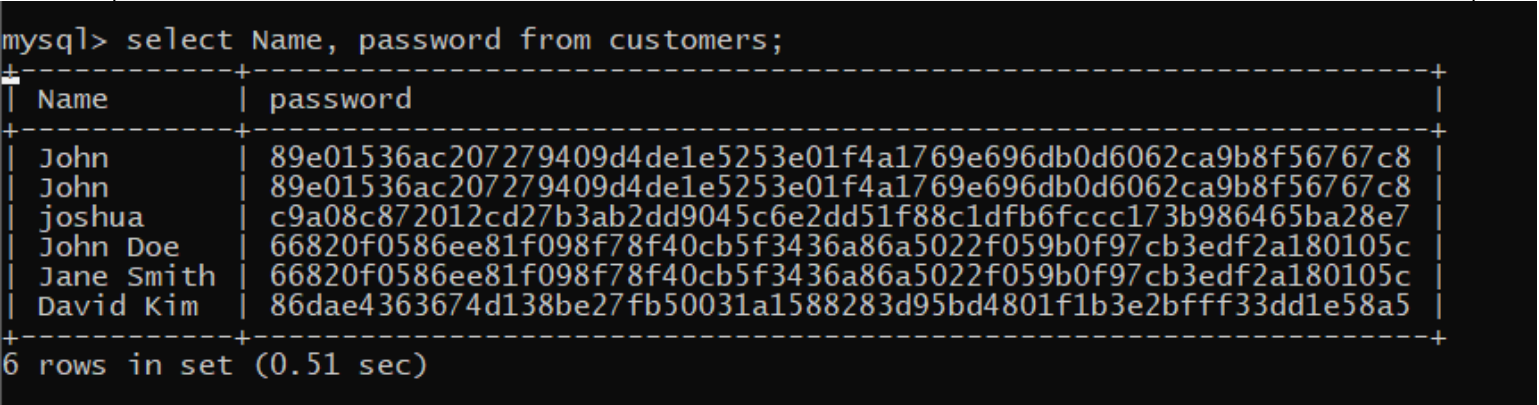


```
mysql> use onlineshopping;
Database changed
mysql> show tables;
+-----+
| Tables_in_onlineshopping |
+-----+
| cart                      |
| categories                |
| customers                 |
| deliveries                |
| orderdetails              |
| orders                    |
| payments                  |
| products                  |
+-----+
8 rows in set (0.38 sec)
```

4. Security: Data Hashing Implementation

I utilized the SHA2 algorithm to encrypt sensitive user passwords. This conversion to a 256-bit hash ensures that the original text is never visible in the database tables.

```
mysql> UPDATE customers SET Password = SHA2('UserPassword123', 256);
mysql> SELECT Name, Password FROM customers;
```



```
mysql> select Name, password from customers;
+-----+-----+
| Name      | password |
+-----+-----+
| John      | 89e01536ac207279409d4de1e5253e01f4a1769e696db0d6062ca9b8f56767c8 |
| John      | 89e01536ac207279409d4de1e5253e01f4a1769e696db0d6062ca9b8f56767c8 |
| joshua    | c9a08c872012cd27b3ab2dd9045c6e2dd51f88c1dfb6fccc173b986465ba28e7 |
| John Doe  | 66820f0586ee81f098f78f40cb5f3436a86a5022f059b0f97cb3edf2a180105c |
| Jane Smith | 66820f0586ee81f098f78f40cb5f3436a86a5022f059b0f97cb3edf2a180105c |
| David Kim | 86dae4363674d138be27fb50031a1588283d95bd4801f1b3e2bfff33dd1e58a5 |
+-----+-----+
6 rows in set (0.51 sec)
```

5. Access Control & Authorization

I configured role-based security by creating specific users and applying the principle of least privilege using GRANT and REVOKE commands.

```
mysql> CREATE USER 'user'@'localhost' IDENTIFIED BY 'password';
mysql> GRANT SELECT, INSERT ON OnlineShopping.* TO 'user'@'localhost';
mysql> REVOKE DELETE ON OnlineShopping.* FROM 'user'@'localhost';
```

6. Verification and Testing

The security model was verified by logging in as the restricted user and attempting a prohibited action.

The system successfully blocked the operation with an Error .

Here I tested this using the mysql workbench. Below is the screenshot of what happens if the customer tries to delete Some data in the database

```
DELETE FROM OnlineShopping.customers WHERE CustomerID = 1;
ERROR 1142 : DELETE command denied to user 'user'@'localhost';
```

The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'Schemas' tab with the 'onlineshopping' database selected. The main window shows 'Query 1' with two statements: 'use onlineshopping;' and 'DELETE FROM OnlineShopping.customers WHERE CustomerID = 1;'. The 'Output' tab is active, showing a table of execution results. The first three rows show successful 'use onlineshopping' actions. The fourth row, corresponding to the DELETE statement, is marked with a red 'x' and shows 'Error Code: 1142. DELETE command denied to user 'user'@'localhost' for t...'. The table has columns for '#', 'Time', 'Action', 'Message', and 'Duration / Fetch'.

#	Time	Action	Message	Duration / Fetch
1	17:59:44	use onlineshopping	0 row(s) affected	0.000 sec
2	18:00:22	use onlineshopping	0 row(s) affected	0.000 sec
3	18:00:24	use onlineshopping	0 row(s) affected	0.000 sec
4	18:01:05	DELETE FROM OnlineShopping.customers WHERE CustomerID = 1	Error Code: 1142. DELETE command denied to user 'user'@'localhost' for t...	0.156 sec